RAHUL MITRA

+65 8515 5413 | mitrarahul2002@gmail.com | linkedin.com/in/rahulmitra-dev | Portfolio | GitHub | **Expected Start Date: July 2027**

EDUCATION

National University of Singapore **Aug 2023 – Jul 2027**
Bachelor of Engineering, Industrial and Systems Engineering (GPA: 4.37/5.00)

- Pursuing a Second Major in Computer Science and a Minor in Mathematics.
- Awarded Certificate of Outstanding Performance in 3D Computer Vision (CS4277): top student in a class of 24, NUS School of Computing.

WORK EXPERIENCE

Warehouse AI & Operations Research Engineer (Part-Time), STMicroelectronics (NUS Capstone) **Aug 2026 – Present**

- Architected a layout-agnostic AI put-away engine ingesting each site's shelf coordinates, dimensions, and operating rules, letting one model adapt to any STMicroelectronics warehouse floor plan without per-site rework.

Global Data Engineer (Operational AI Systems), Part-Time, Abbott Laboratories (Contract) **Jun 2026 – Present**

- Qualified each operating point from read-only Seeq pulls via plant-state classification and 30-minute steadiness checks before any solve, and kept LLM output free of computed numbers through JSON-schema-constrained tool calls and automatic checks of each narration against solver results.

Data Analytics & Product Management Intern, Abbott Laboratories (Converted to Contract) **Jan 2026 – Jun 2026**

- Validated a non-destructive, 15-stage changeover-data pipeline against 5 years of unseen raw data with zero execution failures across operator- and machine-generated anomalies.
- Deployed a team-built APC simulator to production for live internal users on Azure App Service, with Docker packaging, access restrictions, and runtime and session hardening.

RESEARCH EXPERIENCE

Computer Vision Researcher, BlendED x Amazon Robotics (AI+X Global Challenge) **Jul 2026 – Oct 2026**

- Built vision systems turning raw camera feeds into perception-ready frames across 3 workstreams: camera ISP, super-resolution, and image restoration under real-world blur.

PROJECTS

Team Lead & Finalist, Swarmline: SG Defence Tech Hackathon (1,300+ applicants) **Sep 2026**

- Implemented a deterministic Unreal Engine 5.8 C++ simulation with Claude Code; under a ground-link jam, 30 drones confirmed all 8 walking targets in 10 of 10 seeded runs, versus 2.1 of 8 on average for an operator-run fleet.

Finalist, Volt Pulse SG: SMU Hack For Cities (Top 8 Finalist of 50+ teams) **Jan 2026**

- Integrated SEALION embeddings, Supabase vector search, RRF routing, and scheduler-agent planning into an autonomous loop managing electricity and utility savings across multiple HDB households at once.

Competitor, DSTA BrainHack 2025 CTF **Feb 2025**

- Solved web, crypto, reverse-engineering, and systems-security challenges under time pressure, ranking top 25% of 400+ teams.

CO-CURRICULAR ACTIVITIES AND VOLUNTEER EXPERIENCES

Independent Bug Bounty Researcher, YesWeHack **May 2024 – Present**

- Assessed GovTech GBBP12/13, the Land Transport Authority, and 13+ other programs for SSRF, CSRF, SQL/NoSQL injection, authentication bypass, and access-control weaknesses.
- Automated endpoint discovery, payload delivery, and evidence collection for responsible-disclosure reports with Burp Suite, Wireshark, Python, Bash, and custom fuzzing scripts.
- Applied Burp Suite for MitM HTTP interception and Wireshark for OSI Layer 3-4 deep packet inspection and TLS handshake analysis.

Elderly Care Volunteer, NTUC Health Active Ageing Centre **Sep 2025 – Sep 2026**

- Coordinated weekly senior-engagement activities, including home visits to check on residents' wellbeing, hear life stories, and help with everyday tasks, building empathy for accessible, community-focused product design.

NATIONAL SERVICE

Republic of Singapore Navy, Naval Chef & Base Support Assistant **Jan 2021 – Jan 2023**

- Compressed a month-long manual analysis of 6 months of food-waste data into a 3-minute Excel VBA reporting workflow inside a security-constrained environment.

SKILLS AND CERTIFICATIONS

Technical: Burp Suite, Wireshark, SSRF, CSRF, SQL/NoSQL injection, authentication bypass, OAuth2/JWT, OWASP Top 10, fuzzing, reverse engineering, custom exploit scripting, Python, Bash, Rust, Claude Code, ChatGPT, Gemini, ComfyUI
Certifications: Reverse Engineering Windows Executables (digital forensics); Docker for the Absolute Beginner (Stone River eLearning)